Research report: adding rehab data to sports 3D data lets single-camera 3D motion capture beat the Kinect on rehab movements

2026-10-31 · Sonny (AI application team) · status: exploratory, 3 seeds × 3 participant folds · 한국어: RESEARCH_REPORT_KR.md

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Conclusion (first page)

> **A 2D→3D model trained on sports data (ASPset-510) alone could not handle rehab movements. After post-hoc camera calibration of a rehab dataset (KERAAL) and mixing it into training, a single ordinary camera produced more accurate joint angles than a Kinect depth sensor on new participants never used in training.** This holds when training and testing share the same room and camera (§5.5).

	ASPset only	ASPset + KERAAL	Kinect sensor
KERAAL joint-position error (MPJPE)	222.9 ± 3.0 mm	**69.5 ± 0.8 mm**	–
KERAAL joint-angle correlation (r0)	0.59	**0.76**	0.67
KERAAL joint-angle error (RMSE)	20.3°	**13.9°**	18.7°
Trunk-rotation exercise (RTK) angle correlation	0.23 (fails)	**0.72**	0.63
ASPset test MPJPE (original domain)	83.4 mm	**81.1 mm** (no loss)	–

Values are means ± SD over 3 seeds; KERAAL values are measured only on participants never used in training.

Five key findings 1. **The problem was pose variety, not data volume.** Going from 3 to 13 ASPset training subjects left the KERAAL error unchanged. 89% of KERAAL poses are seated, against 13% in ASPset. 2. **The calibration-free KERAAL dataset can be calibrated post hoc.** Using the Kinect as a bridge, Vicon was moved into camera coordinates: 25 mm 3D residual and 6.2 px reprojection error on held-out recordings. 3. **Merged training cuts position error to about a third.** 222.9 → 69.5 mm, better in all 9 seed × participant runs. 4. **It beats the Kinect on angles too.** This angle metric is independent of the calibration. r0 is higher and RMSE lower than the Kinect in all 3 participants, though the model was trained in the same room with the same camera (§5.5). 5. **The RTK failure came from missing data, not a single-camera limit.** After merging, r0 rose from 0.23 to 0.72, past the Kinect (0.63).

Caveat: only 3 test participants (healthy, acted errors), filmed in the same room with the same camera as the training data. Real patients and other locations are not yet tested (§7).

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1. Background and questions

- **Goal:** turn single-camera 2D video into 3D motion-capture data, for use in rehab exercise assessment.
- **Earlier results (10/26–10/30):**
- A model trained on ASPset-510 reached 49 mm PA-MPJPE within ASPset, on par with the paper's baselines.
- On KERAAL, however, angle RMSE was 20°, similar to the Kinect, and it failed on the RTK exercise.
- Training on 3/6/9/13 subjects left KERAAL performance unchanged.
- **Questions:**
- Q1. Can KERAAL get 3D ground truth in the same format as ASPset? (camera calibration)
- Q2. Does merged training help on new participants' rehab movements, and does it keep performance in the original domain (ASPset)?
- Q3. Does it matter whether Vicon or the Kinect is used as ground truth?

2. Data

	ASPset-510	**KERAAL Group 3**
Licence	CC0	CC BY-NC-SA (no raw or derived files in public)
Setting	outdoor, standing full-body sports	indoor, 3 seated arm rehab exercises (CTK, ELK, RTK)
People	17 (train 13 / val 2 / test 2)	3 (healthy, acting errors), 540 recordings
Video	4K 50 fps, 3 cameras	960×540 30 fps, 1 camera (Kinect v2 colour)
3D GT	markerless triangulation + bundle adjustment	Vicon optical markers at 60 fps, 17 rigid targets (position + quaternion)
Calibration	yes (§3.1)	**none → built by us (§3.2)**
Pose range (measured)	13% seated	89% seated; 62% of frames have no similar ASPset pose

3. Camera calibration methods

3.1 ASPset: the authors' method (Nibali et al. 2021, §4)

1. **Capture:** 3 × Canon EOS C200, 24 mm lenses, 4K 50 fps, about 30° apart around the stage. Synchronised by hand on a light being switched off.
2. **Initial calibration:** - An **asymmetric circle-grid board** was moved around the stage with twists and tilts. - **OpenCV** detected the circles to give K per camera, stereo calibration gave E, and gradient descent then reduced the reprojection error of all three cameras at once. - **Lens distortion was assumed zero.** - Detections whose circle rows/columns were not straight (MSE > 0.05) were dropped, and only frames at least 50 px apart were kept (~100).
3. **Bundle adjustment:** - Camera parameters were optimised jointly with 2D joint detections (Stacked Hourglass fine-tuned on human annotations). - It used 5,000 frames with score U < 10 px, which tolerates a constant offset across joints.
4. **Coordinates:** world = first (left) camera; mm; x right · y down · z forward. Each subject has its own calibration.
5. **Limitation:** no GT accuracy in mm is reported.

3.2 KERAAL: our post-hoc calibration (`keraal_calib.py`)

With no calibration board, we used the **Kinect as a bridge**. The structure is the same as ASPset: an initial estimate, then refinement with correspondences.

```
Vicon targets ──② fixed offsets──▶ Vicon joint points ──③ R, t──▶ camera coords ──① K──▶ pixels
                                                          ▲
                                                Kinect joints (camera coords)
```

① **Intrinsics K** - Basis: - KERAAL video at 960×540 is exactly half of the Kinect v2 colour stream (1920×1080). - Kinect file rows match video frames 1:1. - Kinect joints are already stored in camera coordinates (m). - Method: - A pinhole model. From 12 joint correspondences between Kinect joints and YOLO 2D (confidence > 0.5), we fitted fx, fy, cx, cy together with a colour–depth sensor offset t0. - soft-L1 robust least squares. - Kinect axes (x left, y up) were flipped to OpenCV axes (x right, y down). - Result: **fx 502.3, fy 521.4, cx 476.0, cy 256.5, t0 = (4.8, −10.2, −19.6) cm**. Distortion assumed zero (same as ASPset).

② **Vicon targets → joint points** - The 17 Vicon items are rigid targets on body segments, not joint centres. Matching them directly to the Kinect left about 12 cm of residual. - So joint point = target position + target rotation × **fixed offset**, where the offset is a constant in the target's local frame.

③ **Vicon → camera rigid transform** - Kinect joint ≈ R·(joint point) + t, with scale fixed in mm. - R, t (Kabsch) and the offsets (closed-form median) were solved by **alternating least squares**, 30 iterations. Halfway through, frames with Kinect tracking failures (residual above 3× the median) were dropped once. - Vicon was decimated from 60 to 30 fps, and a time lag of ±15 frames was searched per recording. - **Session = participant × exercise.** With one session per participant, only P2 failed (111 mm); per exercise it gave 25–33 mm. - Output: Vicon and Kinect joints per recording, **in ASPset `aspset_17j` order and camera mm**. head_top has no counterpart and is left empty.

Calibration check (fitted on even-numbered recordings, checked on the 270 odd-numbered ones)

Check	Median
Vicon joint points ↔ Kinect joints (3D)	**25.0 mm** (P1 22, P2 29, P3 23)
Vicon projection ↔ YOLO 2D	**6.2 px** (5.9 px with the ASPset-style U score)
Kinect projection ↔ YOLO 2D	6.8 px
Offset sizes	mostly 80–230 mm
Visual check	Vicon points sit on the body's joints in held-out frames

4. Experiment design

Item	Setting
Input	YOLO11x-pose 2D (COCO 17) + confidence, mid-hip + body-extent normalisation
Model	temporal conv net (VideoPose3D-style), 81-frame receptive field, output pelvis-relative 3D for 17 joints
Split	**leave-one-KERAAL-participant-out, 3 folds.** 1/5 of the training participants' recordings held out as KERAAL validation. The test participant is used only at the end

Item	Setting
Model selection	epoch with the lowest (ASPset val + KERAAL val MPJPE) / 2
Loss	per-joint masked MPJPE; joints without GT are left out
Conditions	① ASPset only ② ASPset + KERAAL (Vicon GT) ③ ASPset + KERAAL (Kinect GT) ④ KERAAL only (Vicon GT, 457 epochs for the same number of updates)
Training	AdamW lr 1e-3, 120 epochs, left-right flip augmentation + flip TTA, **seeds 0, 1, 2**
Metrics	**Position:** MPJPE and PA-MPJPE against Vicon joint points (16 joints, no head_top). **Angles:** correlation r0 and RMSE against Vicon **quaternion** relative angles; the angle metric is independent of the calibration. **ASPset test:** checks the original domain is kept
Comparison	Direction and size across the 3 participants (seed-averaged); no p-values

5. Results

5.1 Main results (mean ± SD over 3 seeds, averaged over the 3 participant folds)

Training data	KERAAL MPJPE	PA-MPJPE	Angle r0	Angle RMSE (°)	RTK r0	ASPset MPJPE / PA
① ASPset only	222.9 ± 3.0	102.9 ± 0.1	0.594 ± 0.017	20.3 ± 1.4	0.233	83.4 / 49.4
② **ASPset + KERAAL (Vicon)**	**69.5 ± 0.8**	**52.2 ± 0.2**	**0.764 ± 0.005**	**13.9 ± 0.3**	**0.724**	**81.1 / 48.5**
③ ASPset + KERAAL (Kinect)	77.7 ± 0.2	56.9 ± 0.2	0.712 ± 0.005	15.7 ± 0.4	0.616	81.9 / 48.8
④ KERAAL only (Vicon)	68.4 ± 0.5	52.8 ± 0.5	0.746 ± 0.004	13.6 ± 0.1	0.739	315.4 / 212.1
Kinect sensor	–	–	*0.670*	*18.7*	*0.634*	–
Raw 2D angles	–	–	*0.551*	*29.1*	–	–

5.2 Per participant (seed-averaged)

Training data	P1 MPJPE / r0	P2 MPJPE / r0	P3 MPJPE / r0
ASPset only	189.5 / 0.78	234.5 / 0.59	244.5 / 0.41
ASPset + KERAAL (Vicon)	82.4 / 0.88	70.2 / 0.72	55.9 / 0.69
KERAAL only	92.5 / 0.84	60.3 / 0.72	52.3 / 0.68

5.3 Position MPJPE per exercise (mm)

Training data	CTK	ELK	RTK
ASPset only	230.6	208.7	228.8
ASPset + KERAAL (Vicon)	70.3	61.1	77.5
ASPset + KERAAL (Kinect)	74.4	74.0	85.0

Training data	CTK	ELK	RTK
KERAAL only	70.7	58.3	76.3

5.4 Per-joint MPJPE (mm, ASPset only → ASPset + KERAAL Vicon)

Joint	Before → after	Joint	Before → after
Ankle R / L	431 → **58** / 434 → **65**	Wrist R / L	298 → 147 / 287 → 150
Knee R / L	376 → **55** / 360 → **75**	Elbow R / L	212 → 102 / 212 → 104
Hip R / L	83 → 12 / 83 → 11	Shoulder R / L	169 → 81 / 174 → 74
Head / neck / spine	154 → 70 / 170 → 67 / 120 → 44		

5.5 Participant-level comparison (3 participants, seed-averaged)

The independent unit is the 3 participants. The 9 participant × exercise cells are three exercises from each person and are not independent, so the earlier cell-level p-values (p = 0.004 etc.) have been **removed**. Three participants cannot give a meaningful p-value, so only direction and size are reported. Values are mean differences of per-participant medians.

Comparison	r0 diff, participants better	RMSE diff, participants better
② vs ① ASPset only	+0.170, **3/3**	−6.4°, **3/3**
② vs Kinect	+0.094, **3/3**	−4.8°, **3/3**
③ vs Kinect	+0.042, 3/3	−3.0°, 3/3
④ vs Kinect	+0.077, 3/3	−5.1°, 3/3
② Vicon GT vs ③ Kinect GT	+0.053, 3/3	−1.8°, 3/3
② merged vs ④ KERAAL only	+0.018, 3/3	+0.3°, 0/3

Participant	② r0 / Kinect r0	② RMSE / Kinect RMSE
P1	0.878 / 0.782	13.7° / 19.6°
P2	0.722 / 0.639	14.0° / 17.5°
P3	0.693 / 0.588	14.0° / 19.0°

Scope of the Kinect comparison. The angle computation and reference are the same: same joint-angle function, same Vicon reference, same sync search. These things are not the same: - Our model was trained on other people in the **same room with the same camera** as the test person. The Kinect was not tuned to this setting. So the result means "trained for this setting, it beats the Kinect", not that it would in a setting it has never seen. - Frames where Kinect tracking failed are dropped from the Kinect values only. - The trunk axis of the shoulder angle is SpineShoulder–SpineBase for the Kinect and neck–pelvis for our model.

In all 9 seed × participant runs, ②, ③ and ④ had lower position error than ①.

6. Interpretation: why it worked

1. **It filled the missing poses.** The lower body improved most: ankle 431 → 58 mm, knee 376 → 55 mm. The ASPset-only model predicted seated legs as if standing, which matches the measured pose range (13% vs 89% seated).
2. **The RTK failure was a data problem.** r0 rose from 0.23 to 0.72. The earlier reading "rotation in depth = single-camera limit" was wrong.
3. **The GT format matched ASPset exactly.** The §3.2 calibration put KERAAL GT into the same frame (OpenCV axes, mm, pelvis-relative, same 17-joint order), so the data could be mixed without changing the model or the loss.
4. **Calibration error is below model error.** Calibration 25 mm vs model 69 mm, so GT noise did not spoil training.
5. **The two datasets protect each other.** - On KERAAL, ② and ④ differ little (r0 +0.018, RMSE +0.3°). - But ④ collapsed on ASPset (315 mm), while ② even improved on ASPset (83.4 → 81.1 mm). - So **one model for both domains needs the merge**. The paper's §6.2 (MPI-INF-3DHP + ASPset) shows the same effect.
6. **GT quality matters.** Vicon GT (②) beat Kinect GT (③) even on the calibration-independent angle r0 in all 3 participants (+0.053), so §3.2 steps ②③ (moving Vicon into the camera frame) were worth doing.
7. **The results are stable.** Between-seed SD is 0.8 mm for MPJPE and 0.005 for r0, and all 9 seed × participant runs improved.

7. Limitations, failures, and why

1. **P2 failed in the first calibration (111 mm).** - P2's camera or Vicon frame apparently changed between exercises; per-exercise calibration fixed it. - Whether the camera actually moved is unchecked. P2 also has many Vicon dropouts (42 recordings over 20% missing).
2. **The ASPset-only position error (223 mm) includes a joint-definition mismatch.** - KERAAL GT follows the Kinect definition because the offsets were fitted to Kinect joints; ASPset follows anatomical annotation. - So part of the position gain is "learning the definition". The angle metric is definition-free and also improved a lot, so the conclusion holds.
3. **Wrist error is still large (~150 mm).** Wrist GT is a forearm-target offset fitted to the Kinect wrist, whose tracking is unstable. Functional joint centres (SCoRE) or the hand targets could fix this.
4. **A KERAAL-only model does not generalise.** Two people, indoors, three exercises — too narrow a pose range.
5. **The sample is small.** 3 test participants, too few for a statistical test.
6. **Same room, same camera.** We showed generalisation to new people, not to new places or cameras. The Kinect comparison holds only under this condition (§5.5).
7. **Only acted errors were tested.** Real patients (1A) may move differently.
8. **ASPset's own GT accuracy is unreported.** The paper gives no validation in mm.

8. Next steps

Order	Task	Purpose
1	Apply the merged model to 1A (patients) and 2A (healthy) videos and compare angles with the Kinect	generalisation to real patients and other sessions (no Vicon → angles only)

Order	Task	Purpose
2	Rebuild GT with functional joint centres	better wrists, no dependence on the Kinect definition
3	Rerun error detection (RQ4) with the merged model	link to the application goal
4	External check on OpenCap	different place and camera
5	Joint centres + IK → BVH	complete the "motion-capture data" goal

9. Reproduce

```
python keraal_calib.py --check
python keraal_calib.py E:/KERAAL_data/extracted/group3 E:/KERAAL_mocap3d/det2d E:/KERAAL_mocap3d/calib
python merge_train.py --check
python merge_train.py <ASPset root> <ASPset det> <G3> <G3 det> E:/KERAAL_mocap3d/calib <run> --hold P1 --mode
aspset_vicon --epochs 120 --seed 0
python merge_train.py ... <run> --hold P1 --seed 0 --eval-only E:/ASPset510_data/runs/det_root_s0   # ① ASPset
only
python keraal_eval.py eval <G3> <G3 det> <run> <run>/angles.json
```

- Environment: `E:\mocap3d_env` (Python 3.12, PyTorch, ultralytics), RTX 3090.
- Results: `E:\KERAAL_mocap3d\merge\summary_3seeds.json`, per fold/seed `results.json` and `angles.json`.
- Calibration: `E:\KERAAL_mocap3d\calib\calib.json`.
- **Anything derived from KERAAL (GT, figures) is CC BY-NC-SA and must not go on GitHub.**
- Earlier records: `RESULTS.md` (baseline), `MERGE_REPORT_EN.md` (single-seed first report).

References

- Nibali, Millward, He, Morgan. ASPset: An outdoor sports pose video dataset with 3D keypoint annotations. *Image and Vision Computing* 111 (2021) 104196.
- Nguyen et al. KERAAL dataset. arXiv 2407.00521.
- Pavllo et al. 3D human pose estimation in video with temporal convolutions and semi-supervised training (VideoPose3D). CVPR 2019.